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ETW3482

Week 7 Tutorial



This week's tutorial will have **two sessions**

1. **Quiz for Revision.**
2. **Hands-on practical sessions: Building Decision Tree and Tree-Based Models.**



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Are you ready for the challenge?



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What is the main weakness of the McCulloch-Pitts network?

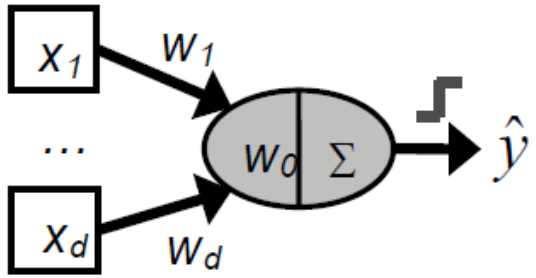
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A Brief History of Neural Networks

- In 1943 McCulloch and Pitts proposed the artificial neural networks.
- The fundamental element of McCulloch-Pitts network is a McCulloch-Pitts neuron.

$$net = w_0 + \sum_{i=1}^{k} w_i x_i$$



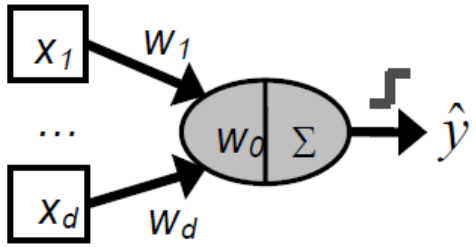
- The result is known as the neuron's net input.
- The McCulloch-Pitts neuron's final output was determined by passing its net input value through a step function, turning each neuron into a linear classifier.

$$f(net) = \begin{cases} 1 \text{ if } net \geq 0 \\ -1 \text{ if } net < 0 \end{cases}$$

A Brief History of Neural Networks

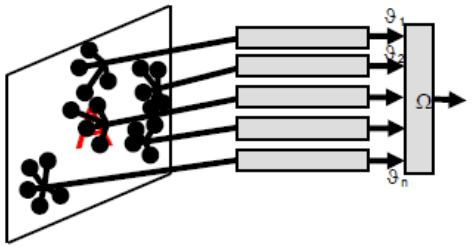
- The weight of the neurons in a McCulloch-Pitts network had to be set manually until Hebb's learning rule was proposed in 1949.

$$\Delta w_{ij} = \eta \hat{y}_i x_j$$



- The η is the neuron's learning rate, where $0 < \eta < 1$.
- Unfortunately, Hebb's learning rule is unstable.
- In 1960, Widrow and Hoff proposed a variant of Hebb's rule called delta rule, which is stable under a range of learning rates.

$$\Delta w_{ij} = \eta (y_i - \hat{y}_i) x_j$$



- In 1959, Frank Rosenblatt implemented a variant of delta rule in a computer model, and he called his learning machine a perceptron.
- He used perceptron for optical character recognition.

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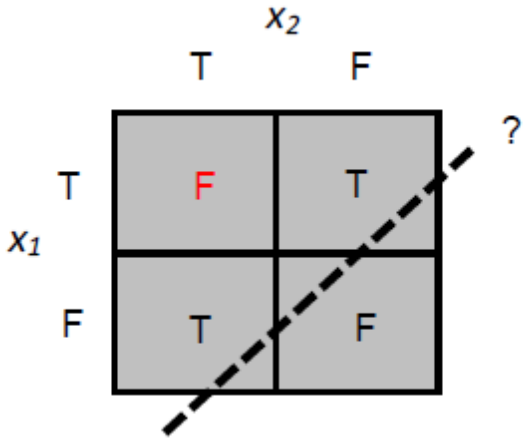
Why can Rosenblatt’s perceptron identify characters A to Z but fail to work on other problems?

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A Brief History of Neural Networks

- Rosenblatt's perceptron able to identify the characters A to Z when presented to an optical scanner but it failed to work on some other problems.
- Minsky and Papert found that Rosenblatt's perceptron could solve only linearly separable problems.
- For example, the simple perceptron couldn't learn and solve the exclusive-or (XOR) logical relationship.
- The reason is that a perceptron that directly connects input neurons to output neurons forms a linear model.



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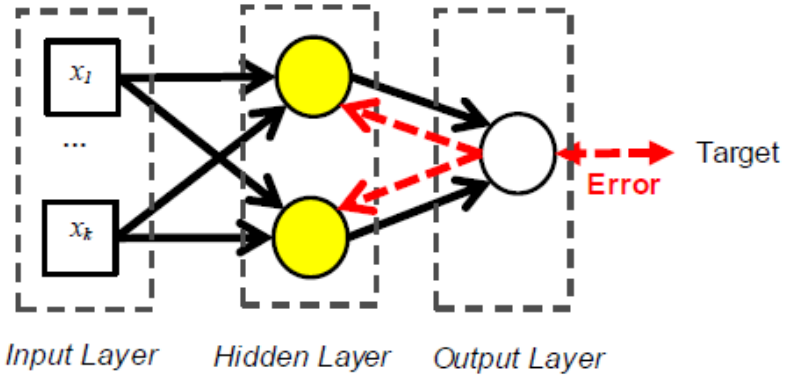
What is the main difference between the Linear Perceptron and Multilayer Perceptron?

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A Brief History of Neural Networks

- Rosenblatt had experimented with multilayer perceptron (MLPs), which the input layer neurons feed into a hidden layer of neurons.
- Adding a hidden layer of neurons was important to generate nonlinear input-output mapping.
- The Widrow-Hoff learning rule failed to estimate weights for MLPs because it needs both input and output values to adjust its connections.
- The generalized delta rule (back propagation) solved this problem by replacing the discontinuous step function with a continuous sigmoid function.



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Select the options that correctly describe the advantages and disadvantages of neural networks.

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Advantages of Neural Networks

- No functional form
  - Neural networks modelling does not need to specify the functional form of the model
  - The models can be constructed when the relationship between the inputs and outputs is unknown.
- Universal approximation
  - Neural networks can model any input-output relationship with a sufficient number of hidden units and enough time.
- One of the fastest scoring models
  - Although neural networks have a slow training times, after they are trained, neural networks can efficiently score large volumes of data.

Disadvantages of Neural Networks

- Lack of interpretability
  - Neural networks are the famous black-box models
  - Neural networks are most appropriate for pure prediction tasks.
- Underperformance
  - Sometimes neural networks do not outperform simpler and easier implement models like regression.
  - Modelers are less likely to construct a neural network model successfully if they are not familiar with the theoretical and practical implications of the neural networks' architecture and options.
- Impact of noisy data
  - Neural networks cannot produce a superior model if the data has a low signal to noise ratio.

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Which of the following are the components of a basic Multilayer Perceptron Neural Network?

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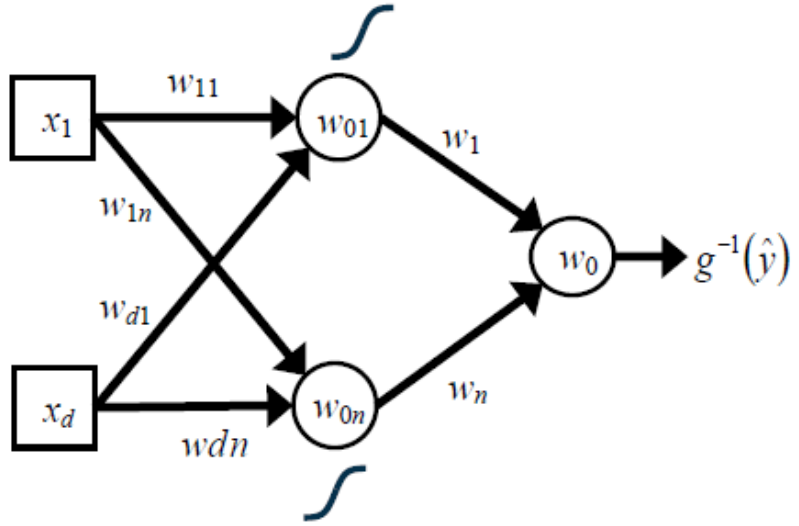
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Multilayer Perceptron

$$g^{-1}(\hat{y}) = w_0 + \sum_{i=1}^{h} w_i g_i \underbrace{\left(w_{0i} + \sum_{j=1}^{d} w_{ij} x_j \right)}_{\text{hidden layer}}$$

There are three layers in the basic multilayer perceptron (MLP) neural network:

1. **An input layer**
   A layer containing a neuron for each input variable without weights.
2. **A hidden layer**
   A layer with hidden units that perform a sigmoidal transformation of the weighted and summed inputs activations.
3. **An output layer**
   A layer that combines the nonlinear hidden layer activation values.



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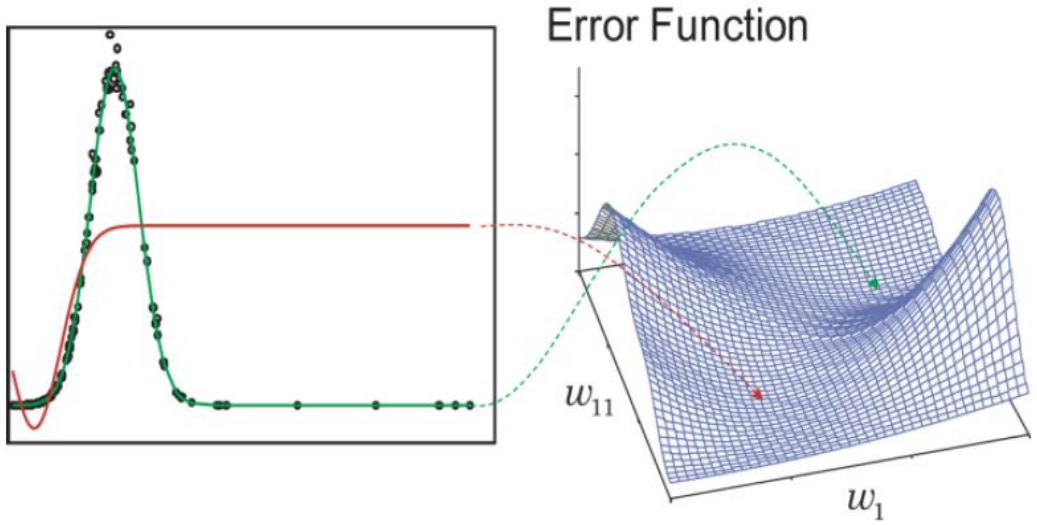


Which of the following is TRUE for the estimation of neural network model weights?

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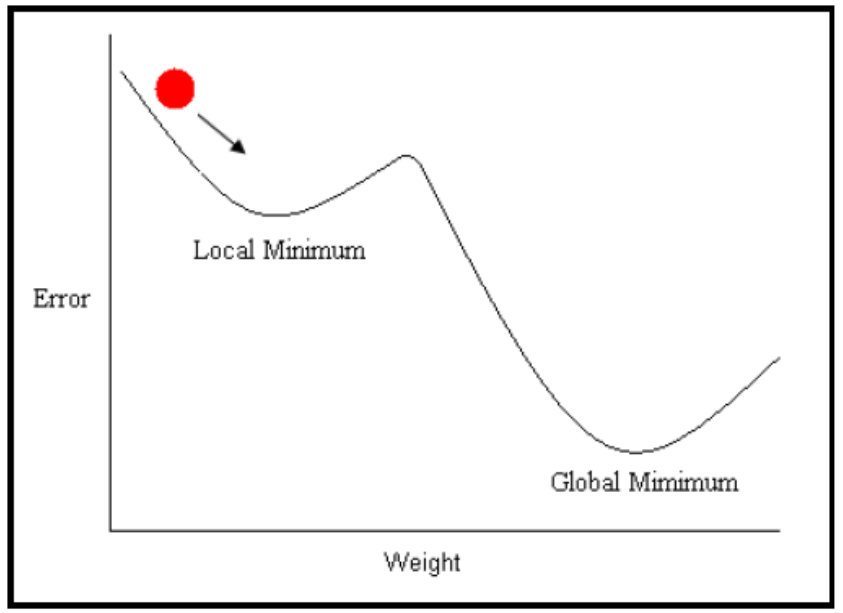
Global Minima and Weights Estimation



- A set of weights are found based on the global minimum of the error function surface.
- The curse of dimensionality makes the brute-force search of the parameter space infeasible.

Global Minima and Weights Estimation

- None of the algorithms can guarantee convergence to a global error minimum.
- The search optimization algorithms are heuristic methods that are vulnerable to local minima on the error surface.
- From the optimization algorithm's perspective, when a local minimum is achieved, any movement away from the bottom leads to an increase in error. Therefore the search stops.



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Refer to the backpropagation algorithm shown below. Which of the following components in the backpropagation algorithm for an interactive updating procedure update vector can reduce the hemstitching movement?

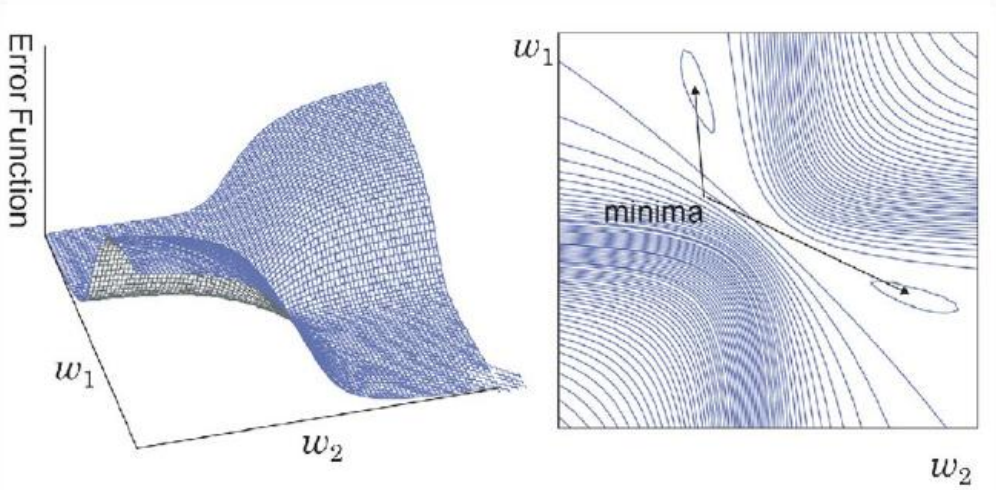
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Iterative Updating

- The heuristic optimisation method is used to search for an error minimum.
- The optimisation use the local features of the error surface to guide their descent.
- The weights associated with a given error minimum are located using the iterative updating procedure:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \delta^{(t)}$$



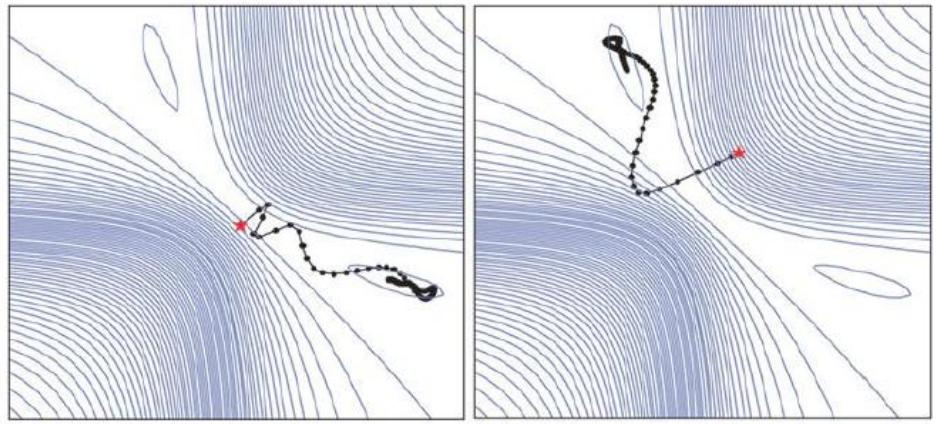
1. Initialise the weight vector to small random values, $\boldsymbol{w}^{(0)}$.
2. Use an optimization method to determine the update vector, $\boldsymbol{\delta}^{(t)}$
3. Add the update vector to the weight values from the previous iteration to generate new estimates, $\boldsymbol{w}^{(t+1)} = \boldsymbol{w}^{(t)} + \boldsymbol{\delta}^{(t)}$.
4. If none of the specified convergence criteria have been achieved, then go to step 2.

Back Propagation

- Initially, the back propagation algorithm used only gradient descent to determine an appropriate set of weights.
- The gradient, $\nabla \boldsymbol{g}^{(t)}$, is the vector of partial derivatives of the error function with respect to the weights.
- By negating the step size parameter, η, a step is made in the direction that is locally steepest downhill.

$$\boldsymbol{\delta}^{(t)} = -\eta \nabla \boldsymbol{g}^{(t)}$$

$$\delta^{(t)} = -\eta \nabla g^{(t)} + \alpha \delta^{(t-1)}$$



87 iterations
(η = 0.5, α = 0.9)

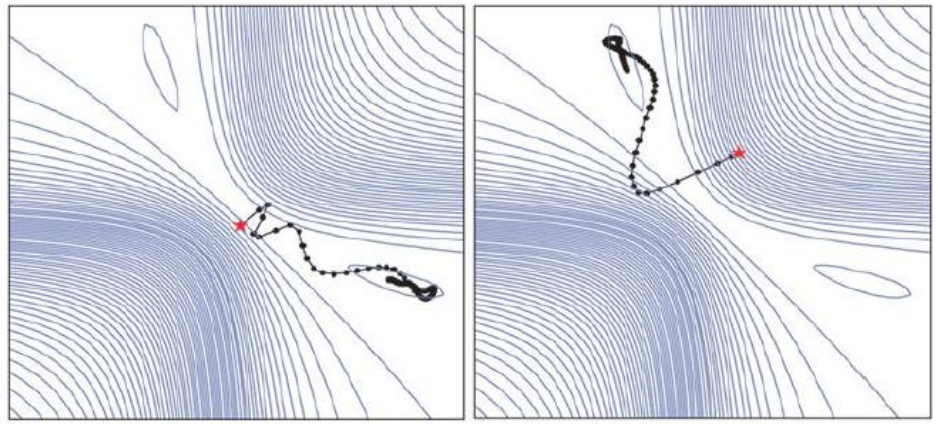
285 iterations
(η = 0.1, α = 0.9)

Back Propagation

- Unfortunately, as the gradient descent approaches the desired weigths, it exhibits numerous back-and –forth movements known as **hemstitching**.
- To avoid the training iterations wasted in this hemstitching, an updated version of back propagation included a **momentum term** α , where $0 \leq \alpha \leq 1$.

$$\boldsymbol{\delta}^{(t)} = -\eta \nabla \boldsymbol{g}^{(t)} + \alpha\, \boldsymbol{\delta}^{(t-1)}$$

$$\delta^{(t)} = -\eta \nabla \mathrm{g}^{(t)} + \alpha \delta^{(t-1)}$$



87 iterations
(η = 0.5, α = 0.9)

285 iterations
(η = 0.1, α = 0.9)

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Audience Q&A

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PART 2:
PRACTICAL HANDS-ON SESSION



Preparation for the practical session

- The Week 7 Demo Guides is uploaded in the Real-time section on Week 7 Moodle.
- Refer to the Week 7 Demo Guides and follow the tutor's instructions.
- Now, let's go through the demo and answer the following questions.

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WRAP UP



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Q&A



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